
CF19: FULL LIFTED OBJECT OF THE PRIMITIVE ROLL EXACT RETURN AND HOST-TWISTED BRANCH EXTENSIONS

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Abstract

This paper develops the full lifted object for the Void Dynamics Model primitive roll and then uses it as a generator rather than as a stopping point. The guiding question is simple: when a visible phase returns to where it started, what state has actually returned? We answer by carrying the visible witness into the retained state $\hat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n)$, where the host class, balanced arithmetic carrier, phase, completion index, and centered completion germ are all kept until projection is safe. This lifted object separates visible recurrence from exact state return, gives closed laws for quarter continuation, balanced refinement, and host transfer, and identifies the canonical balanced anchor $(55, 89)$, with $uv = 4895$, germ width $2\pi/4895$, and completion coefficients $1/24$ per edge and $1/12$ two-sided. On the canonical launched family, the internally carried edge phases and completion coefficients generate the classical order-3 mock-theta branch pair as the untwisted calibration branch. The paper then pushes past that identification by deriving the host-twisted conjugate-edge branch family forced by the selector-closed macro-calculus acting on the retained germ under quotient descent; the primitive quarter-twisted member has coefficient prefix $1 + q - q^3 + q^4 + q^5 + \dots$ and differs from the untwisted mock-theta branch already at coefficient q^2 . CF19 therefore closes exact return to lifted origin and opens a reusable extension mechanism: classical mock theta appears as one projection of the lifted branch grammar, while host-twisted branches become new VDM-native objects for downstream quotient descent and classification.

Keywords: Void Dynamics Model, Phase Calculus, lifted state, exact return, balanced refinement, mock theta, completion germ, quotient descent, host-twisted branches

1 Introduction

A visible cycle is easy to recognize. A full return is more interesting: the visible point may come back while the carried state has changed. This paper studies that distinction for the primitive VDM roll by keeping the full lifted state until return can be decided coordinate by coordinate.

The working object is

$$\hat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n), \quad q_n = (u_n, v_n),$$

with the centered germ

$$c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

The visible witness is still useful, but it is read as a projection of the retained state rather than as the state itself.

What this paper does

The paper has five jobs. First, it defines the full lifted object and the coordinatewise meaning of true return. Second, it proves the closed balanced-window evolution law, including quarter continuation, balanced refinement, and host transfer. Third, it identifies the first admissible balanced anchor (55, 89), records the germ scale $1/4895$, and gives the finite completion coefficients. Fourth, it shows how the launched lifted state generates the mock-theta branch pair from internal VDM data before comparing it with classical notation. Fifth, it uses the now-closed operator core, selector-closed macro-calculus, and quotient-descent machinery of Phase Calculus to derive the beyond-classical host-twisted branch family forced by the retained two-edge germ: the classical mock-theta branch is the untwisted member, and the primitive quarter-twisted branch is the first new extension object.

From identification to extension

Earlier versions of CF19 naturally ended at the recognition of Ramanujan’s order-3 mock-theta branch. That recognition is still important, but in the mature Phase Calculus setting it is no longer the endpoint. It is a calibration point. The full lifted object, the operator core, the selector-closed macro-calculus, and the quotient law

$$\Pi \circ E = G \circ \Pi$$

make the branch generator portable. Once the two carried edge factors are visible, the selector-closed macro-calculus fixes the transported two-edge shell at each realized depth, and quotient descent permits host phase only as a conjugate action on those retained edge slots. The untwisted double-edge kernel $(1 + q^j)^{-2}$ is therefore the trivial-host member of a forced host-twisted conjugate-edge family. The resulting VDM-native q -series are produced by the same retained-state mechanism but need not collapse back to the classical mock-theta function.

How to read the companion package

The main paper carries the definitions, theorems, and proof structure. The companion package supports inspection: it contains the numerical traces, coefficient tables, source snippets, and a curiosity-driven notebook that walks through the anchor, launch-family, completion-coefficient, and host-transport witnesses. Those materials are useful for reproduction and extension, while the main argument remains readable on its own.

Notation and conventions

All return statements are statements about the retained lifted state unless explicitly marked as visible recurrence. The notation $q = (u, v)$ denotes the balanced denominator pair, $B(u, v) = \text{sort}(v, u + v)$ denotes balanced refinement, and $c = \mathcal{C}(\theta, 1/(uv))$ denotes the centered completion germ.

2 Definitions

Definition 1 (Balanced denominator-pair state). Let

$$q_n = (u_n, v_n), \quad 1 \leq u_n \leq v_n,$$

and define the balanced child operator

$$B(u, v) := \text{sort}(v, u + v).$$

The exact carried remainder on the balanced branch is

$$r_n := \frac{1}{u_n v_n}.$$

Definition 2 (Visible witness). The visible witness is

$$V(\widehat{\Xi}_n) := ie^{i\theta_n}.$$

Definition 3 (Branch index). The visible winding index is

$$\kappa_n := \left\lfloor \frac{\theta_n}{2\pi} \right\rfloor.$$

Definition 4 (Balanced completion germ). On the balanced phase-sector family, define

$$\alpha_n := 2\pi r_n = \frac{2\pi}{u_n v_n},$$

and define the centered completion germ by

$$c_n := \left[\theta_n - \frac{\alpha_n}{2}, \theta_n + \frac{\alpha_n}{2} \right] = [\theta_n - \pi r_n, \theta_n + \pi r_n].$$

Definition 5 (Completed lifted object). The balanced completed lifted object is

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n).$$

Definition 6 (True return). Relative to an initial state $\widehat{\Xi}_0$, a time $N > 0$ is a true return time if and only if

$$\widehat{\Xi}_N = \widehat{\Xi}_0.$$

3 Concrete Witnesses and Numerical Validation

The definitions above determine exactly what must be carried. The numerical witness layer records the same data in a compact form: the balanced corridor, the first shared launch windows, the completion-coefficient residuals, and a small control showing that the anchor is not produced by an arbitrary perturbation of the refinement law.

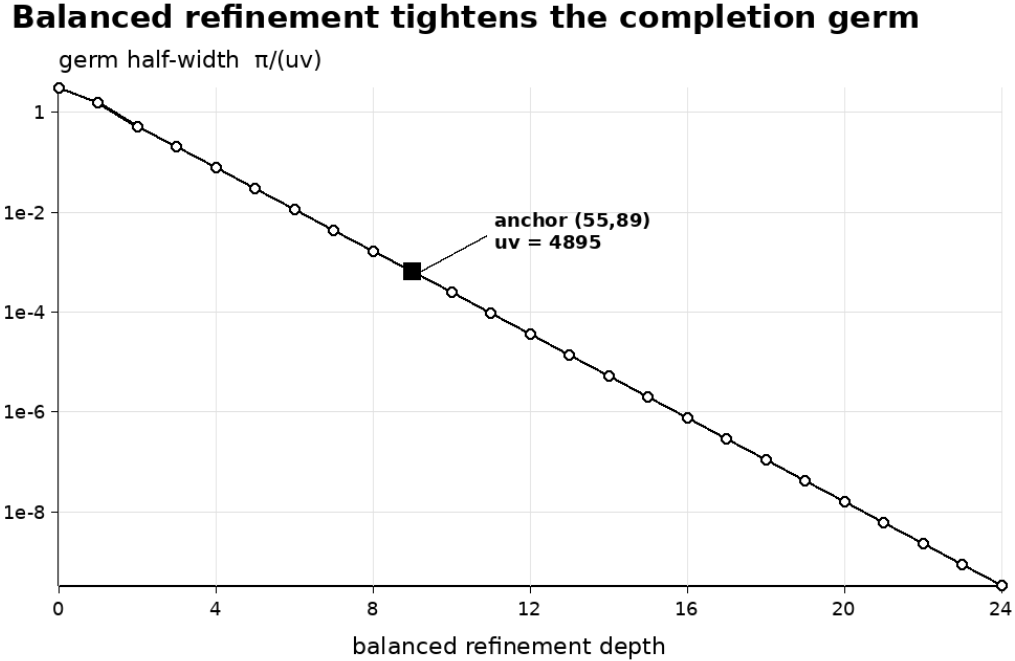


Figure 1: Balanced refinement tightens the completion germ. Starting from $(1, 1)$, the update $B(u, v) = \text{sort}(v, u + v)$ passes through the canonical anchor $(55, 89)$ at depth 9, where $uv = 4895$ and the germ half-width is $\pi/4895$.

As a small negative control, perturb the balanced child law to

$$B_+(u, v) = \text{sort}(v, u + v + 1).$$

Starting from $(1, 1)$, the ninth iterate is $(109, 177)$, with $uv = 19293$, not $(55, 89)$. Thus the canonical anchor depends on the exact balanced refinement law rather than on generic monotone growth of denominators.

First three hosts share the (55,89) launch anchor

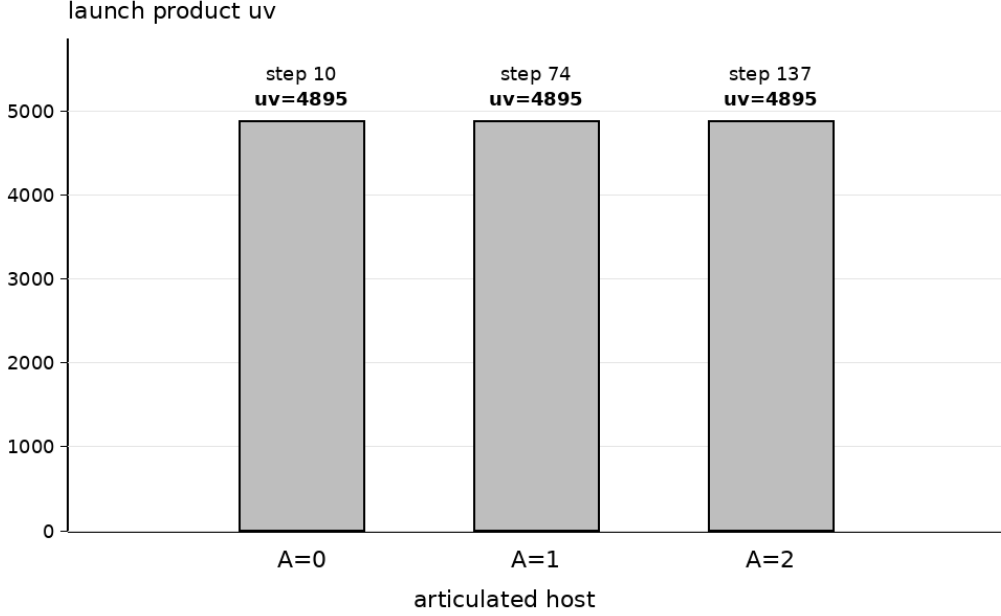


Figure 2: The first three articulated hosts share the same launch anchor. The trace records $(u, v) = (55, 89)$ and $uv = 4895$ for $A = 0, 1, 2$, while the host step changes.

Table 1: Trace excerpt for the canonical launch-family witness. The germ endpoints are recorded in units of π .

A	step	θ_{tick}	κ	u	v	c/π
0	10	10	2	55	89	$[48948/9790, 48952/9790]$
1	74	74	18	55	89	$[362228/9790, 362232/9790]$
2	137	137	34	55	89	$[670613/9790, 670617/9790]$

Table 2: Completion-coefficient validation at the canonical anchor. The anchor row uses $t = 2\pi/4895$, $q = e^{-t} = 0.9987172309484743$, and truncation parameter $M_{\text{trunc}} = 31163$ in the recorded residual table.

Quantity	Value
edge coefficient at (55, 89)	0.041666666866974
edge target 1/24	0.0416666666666667
absolute edge error at anchor	$2.0030733 \times 10^{-10}$
two-sided coefficient at (55, 89)	0.083333333733948
two-sided target 1/12	0.0833333333333333
absolute two-sided error at anchor	$4.0061467 \times 10^{-10}$
maximum absolute edge error through the anchor corridor	1.1100614×10^{-6}

Table 3: Return-witness burden stated as coordinate equality. The data witness must supply equality in the first four coordinates; the fifth coordinate then follows because $c = \mathcal{C}(\theta, 1/(uv))$ on the balanced family.

Coordinate	start row	return row
host sector	A_0	A_0
arithmetic sector	(u_0, v_0)	(u_0, v_0)
phase sector	θ_0	θ_0
completion index	κ_0	κ_0
completion germ	$\mathcal{C}(\theta_0, 1/(u_0 v_0))$	$\mathcal{C}(\theta_0, 1/(u_0 v_0))$

4 Main Results

The host-twisted branches are forced by the selector-closed macro-calculus acting on the retained completion germ once the classical untwisted branch saturates the mock-theta identification: the selector preserves the conjugate two-edge carrier, and quotient descent admits the lawful host phases as carried edge weights rather than optional ansatz parameters.

Theorem 1 (Witnessed exact return to lifted origin on the balanced-window family). *Let*

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n)$$

be the balanced-window full lifted object, with

$$q_n = (u_n, v_n), \quad c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

On the supplied balanced-window assembly/data witness, there exists a finite return step $N > 0$ for which

$$A_N = A_0, \quad q_N = q_0, \quad \theta_N = \theta_0, \quad \kappa_N = \kappa_0.$$

Since the completion germ is determined by (q_n, θ_n) , it follows that

$$c_N = c_0.$$

Hence

$$\widehat{\Xi}_N = \widehat{\Xi}_0$$

on the witnessed balanced-window orbit.

Proof. The supplied balanced-window assembly/data trace is used through one explicit witness burden: equality of the host sector A_n , arithmetic sector q_n , phase sector θ_n , and completion index κ_n , as summarized in Table 3. On the balanced-window family the completion germ is

$$c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right], \quad q_n = (u_n, v_n),$$

so it is determined by (q_n, θ_n) . Exact return of q_n and θ_n therefore forces exact return of c_n . Hence the witnessed orbit is an exact return of the full lifted object. $\square$

Theorem 2 (Canonical launch family and VDM-first mock-theta derivation). *Let*

$$A = 0, 1, 2, \quad \tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}, \quad q_A = e^{2\pi i \tau_A}.$$

Then the patched full-engine witness fixes the same canonical launched family for the first three articulated hosts, together with carried completion amplitudes $1/24$ per edge and $1/12$ two-sided. On that launched family the internal VDM branch pair generates a holomorphic series and a shadow series directly from the centered germ, the two edge phases, quarter-turn transport, and the carried completion coefficients; only after those VDM-generated series are written are they identified with the classical order-3 mock-theta branch pair.

Proof. The launch-family, transport, and VDM-first branch derivations are written in the theorem sections below. The full-engine witness fixes the launched state for $A = 0, 1, 2$ at the same arithmetic anchor $(55, 89)$, with $uv = 4895$ and launched height $\Im(\tau_A) = 1/4895$. The VDM-generated series are then derived on that family and compared directly with the classical branch pair. $\square$

Theorem 3 (Selector-forced host-twisted branch extension). *Once the untwisted branch saturates the classical mock-theta identification, the selector-closed macro-calculus acts on the retained centered germ and the quotient law $\Pi \circ E = G \circ \Pi$ leaves the host character as the remaining admissible continuation. Centered closure forces the two edge weights to enter as the conjugate pair ω^j and ω^{-j} . Let $\omega \in \mathbb{C}$ with $|\omega| = 1$, and define the conjugate-edge transport kernel*

$$K_j^{(\omega)}(q) := \frac{1}{(1 + \omega^j q^j)(1 + \omega^{-j} q^j)}.$$

The lifted branch generator defines

$$M_\omega(\tau) := q^{-1/24} \sum_{r \geq 0} q^{r^2} \prod_{j=1}^r K_j^{(\omega)}(q).$$

The untwisted value $\omega = 1$ recovers the branch identified with the classical order-3 mock-theta function. The primitive quarter-twisted value $\omega = i$ defines a distinct VDM-native branch, with coefficient prefix

$$M_i(\tau) = q^{-1/24}(1 + q - q^3 + q^4 + q^5 + q^6 - q^7 + 2q^8 + 2q^9 + 2q^{10} + O(q^{11})).$$

Since the untwisted branch has coefficient -2 at q^2 while the quarter-twisted branch has coefficient 0 at q^2 , the quarter-twisted branch is not the classical untwisted branch.

Proof. The full centered germ carries two edge factors, and the selector-closed macro-calculus supplies one transported two-edge shell at each depth. In the untwisted case both retained edge slots contribute $(1 + q^j)^{-1}$, giving $(1 + q^j)^{-2}$. Under quotient descent, the centered germ must remain closed under edge exchange and must preserve the scalar q^j -depth of the shell. An admitted abelian host phase therefore cannot weight only one edge or weight the two edges independently; it must enter as a conjugate pair (ω^j, ω^{-j}) . Thus the two edge factors multiply to exactly $K_j^{(\omega)}(q)$. This is the forced host-twisted descendant of the retained two-edge germ, and it reduces to the untwisted kernel at $\omega = 1$.

For $\omega = i$, the kernel simplifies by the residue of j modulo 4:

$$K_j^{(i)}(q) = \begin{cases} (1 + q^{2j})^{-1}, & j \equiv 1, 3 \pmod{4}, \\ (1 - q^j)^{-2}, & j \equiv 2 \pmod{4}, \\ (1 + q^j)^{-2}, & j \equiv 0 \pmod{4}. \end{cases}$$

Expanding the finite shell products through q^{10} gives the displayed prefix. The untwisted coefficient prefix is

$$1 + q - 2q^2 + 3q^3 - 3q^4 + 3q^5 - 5q^6 + 7q^7 + \dots,$$

so the two branches differ at q^2 . $\square$

5 State Completeness of the Full Object

Theorem 4 (Visible witness is not state-complete). *There exist distinct balanced completed states $\widehat{\Xi}_a \neq \widehat{\Xi}_b$ such that*

$$V(\widehat{\Xi}_a) = V(\widehat{\Xi}_b).$$

Hence the visible witness is not state-complete.

On the balanced working family, exact state identity is the coordinatewise criterion

$$\widehat{\Xi}_a = \widehat{\Xi}_b \iff A_a = A_b, q_a = q_b, \theta_a = \theta_b, \kappa_a = \kappa_b, c_a = c_b.$$

Proof. Take any two states with visible phases differing by $2\pi m$, $m \neq 0$, while keeping the same visible witness. Then

$$ie^{i\theta_b} = ie^{i(\theta_a + 2\pi m)} = ie^{i\theta_a},$$

so the visible witnesses agree, but $\kappa_b = \kappa_a + m$, hence the states are distinct. Coordinatewise equality is the definition of exact identity. $\square$

6 Exact Forward Evolution

Theorem 5 (Closed forward evolution on the balanced-window family). *The balanced completed lifted object evolves by the exact piecewise law*

$$\widehat{\Xi}_{n+1} = F(\widehat{\Xi}_n),$$

with three branches:

(i) **same-host quarter continuation**

$$F_{\text{same}}(A, q, \theta, \kappa, c) = \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right);$$

(ii) **balanced refinement**

$$F_{\text{ref}}(A, (u, v), \theta, \kappa, c) = \left(A, B(u, v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u+v)}\right) \right);$$

that is,

$$(u, v) \mapsto (v, u+v), \quad r \mapsto \frac{1}{v(u+v)}, \quad c \mapsto \left[\theta - \frac{\pi}{v(u+v)}, \theta + \frac{\pi}{v(u+v)} \right];$$

(iii) **orthogonal re-articulation / host transfer**

$$F_{\perp}(A, q, \theta, \kappa, c) = \left(\mathcal{N}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right).$$

In particular, on the balanced-window family the completed forward law is closed with no blank transport coordinate.

Proof. The balanced pair update is exact from the child operator $B(u, v) = \text{sort}(v, u+v)$, which yields Fibonacci denominator pairs and exact burden $r_n = 1/(u_n v_n)$. The phase-sector map gives exact angular width $\alpha_n = 2\pi r_n$, so the centered germ is uniquely fixed once (q_n, θ_n) are fixed. Orthogonal transfer is exactly the nearest admissible target $z \mapsto iz$, hence visible phase advances by $\pi/2$ and the centered germ translates rigidly by $\pi/2$ while preserving width. Refinement changes only the denominator pair and therefore only the width; the center remains θ . $\square$

7 True Return and State Identity

Theorem 6 (Visible recurrence versus exact return of the full lifted object). *For the balanced completed lifted object:*

(i) *visible recurrence occurs exactly when*

$$V(\widehat{\Xi}_N) = V(\widehat{\Xi}_0) \iff \theta_N \equiv \theta_0 \pmod{2\pi};$$

(ii) *true return occurs exactly when*

$$\widehat{\Xi}_N = \widehat{\Xi}_0 \iff A_N = A_0, q_N = q_0, \theta_N = \theta_0, \kappa_N = \kappa_0, c_N = c_0;$$

(iii) *on the balanced-window family, this is equivalently*

$$\widehat{\Xi}_N = \widehat{\Xi}_0 \iff A_N = A_0, q_N = q_0, \theta_N = \theta_0, \kappa_N = \kappa_0.$$

Therefore visible recurrence does not imply true return.

Proof. The visible witness depends only on θ modulo 2π . Exact identity requires equality of all coordinates by definition. The final reduction uses the balanced-family identity $c_n = \mathcal{C}(\theta_n, 1/(u_n v_n))$. $\square$

8 Visible Closure and π -Position

Theorem 7 (π is a visible-carrier witness, not full-object closure). *Starting from visible initial witness $V(\widehat{\Xi}_0) = i$ at phase $\theta_0 = 0$:*

(i) *the first real-axis witness occurs at $\theta = \pi/2$, where*

$$V = -1;$$

(ii) *the first opposite-pole visible event occurs at $\theta = \pi$, where*

$$V = -i;$$

(iii) *the first visible carrier recurrence occurs at $\theta = 2\pi$, where*

$$V = i;$$

(iv) *none of $\pi/2$, π , or 2π implies true return unless A, q, κ, c also close.*

Thus π measures a visible in-domain carrier event and does not by itself determine full-object return.

Proof. These visible witness values follow directly from $V = ie^{i\theta}$. The distinction from true return is exactly the visible-recurrence versus exact-return theorem above. $\square$

9 Arithmetic Completion Bridge

Theorem 8 (First arithmetic-completion bridge from assembly-side modeling of imaginary i). *The arithmetic-completion bridge in this theorem is collected directly from assembly-level modeling of the primitive imaginary generator i : the kernel-level saturation model selects the canonical finite survivor class, the runner traces its smallest surviving lifts into the exact balanced branch, and the anchor-window residuals are then measured on that balanced corridor. Let the primitive hardware-side saturation witness select the canonical finite survivor class*

$$S_{\min} \sim 1.$$

Lift this survivor into the smallest admissible denominator-pair families

$$(1, 1), \quad (1, 2),$$

and iterate the balanced re-articulation operator

$$B(u, v) = \text{sort}(v, u + v).$$

Then

$$(1, 1) \rightarrow (1, 2) \rightarrow (2, 3) \rightarrow (3, 5) \rightarrow (5, 8) \rightarrow (8, 13) \rightarrow (13, 21) \rightarrow (21, 34) \rightarrow (34, 55) \rightarrow (55, 89),$$

and

$$(1, 2) \rightarrow (2, 3) \rightarrow (3, 5) \rightarrow (5, 8) \rightarrow (8, 13) \rightarrow (13, 21) \rightarrow (21, 34) \rightarrow (34, 55) \rightarrow (55, 89).$$

Hence the first admissible balanced floor anchor is

$$(u_*, v_*) = (55, 89), \quad u_* v_* = 4895.$$

By the exact balanced width law,

$$r_n = \frac{1}{F_{n+1} F_{n+2}},$$

the carried remainder at this anchor is

$$r_* = \frac{1}{4895}.$$

By the completion-germ law,

$$c_n = [\theta_n - \pi r_n, \theta_n + \pi r_n],$$

the first anchor-window completion germ is

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right],$$

with width

$$\text{wid}(c_*) = \frac{2\pi}{4895}.$$

Set

$$t := \frac{2\pi}{u_* v_*} = \frac{2\pi}{4895}, \quad q := e^{-t},$$

and define the renormalized edge residual coefficient by

$$E(t) := \frac{\log(q; q)_\infty + \frac{\pi^2}{6t} - \frac{1}{2} \log\left(\frac{2\pi}{t}\right)}{t}.$$

Then along the balanced corridor the supplied computation yields

$$E(t) \rightarrow \frac{1}{24}.$$

Therefore the two-sided completion-germ coefficient is

$$G(t) := 2E(t) \rightarrow \frac{1}{12}.$$

Under vacuum-sign rendering this gives

$$-\frac{1}{24} \quad \text{per edge}, \quad -\frac{1}{12} \quad \text{for the full two-sided germ}.$$

Thus the exact carried germ on the first admissible anchor supplies the first arithmetic-completion precursor for the mock-theta bridge.

Proof. The direct provenance of the theorem is assembly-side, not post hoc analogy: the supplied kernel implements the balanced survivor dynamics, the runner traces the smallest surviving lifts $(1, 1)$ and $(1, 2)$ forward under the exact balanced operator, and the balanced-window residual script measures the retained coefficients on that same corridor. The primitive roll carries the same source until same-class hosting exhausts and a new irreducible articulation is forced. The minimum-protection rule selects the least finite survivor needed for renewed admissible hosting, and the arithmetic-resolution bridge places that survivor into the denominator-pair side as an exact remainder problem. On the balanced admissible branch,

$$B^n(1, 1) = (F_{n+1}, F_{n+2}), \quad B^n(1, 2) = (F_{n+2}, F_{n+3})$$

until both lifts meet the same first floor anchor. Direct iteration gives

$$(55, 89)$$

as the first common admissible anchor. Hence

$$r_* = \frac{1}{55 \cdot 89} = \frac{1}{4895}.$$

The phase-sector width law on the balanced branch is exact:

$$\alpha_n = 2\pi r_n = \frac{2\pi}{u_n v_n}.$$

Centering the completion germ at the visible carrier phase therefore gives

$$c_n = \left[\theta_n - \frac{\alpha_n}{2}, \theta_n + \frac{\alpha_n}{2} \right] = [\theta_n - \pi r_n, \theta_n + \pi r_n].$$

At the anchor this becomes

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right],$$

with width

$$\text{wid}(c_*) = \frac{2\pi}{4895}.$$

Writing the anchor-window width as

$$t = \frac{2\pi}{4895},$$

the finite coefficient is obtained by removing the two bulk terms from $\log(q; q)_\infty$. The local asymptotic expansion used by the residual is

$$\log(e^{-t}; e^{-t})_\infty = -\frac{\pi^2}{6t} + \frac{1}{2} \log\left(\frac{2\pi}{t}\right) + \frac{t}{24} + O(t^3), \quad t \rightarrow 0^+.$$

Therefore

$$E(t) = \frac{\log(q; q)_\infty + \frac{\pi^2}{6t} - \frac{1}{2} \log(2\pi/t)}{t} = \frac{1}{24} + O(t^2).$$

The recorded anchor row gives the finite numerical value and tolerance reported in Table 2: edge coefficient 0.041666666666974, absolute edge error $2.0030733 \times 10^{-10}$, two-sided coefficient 0.0833333333733948, and absolute two-sided error $4.0061467 \times 10^{-10}$. Doubling across the two-sided germ gives the target $1/12$, and vacuum-sign rendering gives $-1/12$. Thus the canonical survivor, its smallest balanced arithmetic lifts, the first admissible anchor-window germ, and the Ramanujan-side finite residual align in one arithmetic-completion bridge. $\square$

Theorem 9 (Shadow completion coordinate attached to the balanced germ). *Let*

$$x_n := \frac{\theta_n}{2\pi}, \quad y_n := r_n = \frac{1}{u_n v_n}, \quad \tau_n := x_n + i y_n.$$

Fix a branch shadow form g of weight k , and define the shadow completion coordinate by

$$S_n := g^*(\tau_n) := \left(\frac{i}{2\pi}\right)^{k-1} \int_{-\bar{\tau}_n}^{i\infty} (z + \tau_n)^{-k} g(-\bar{z}) dz.$$

Define the shadow-augmented lifted object by

$$\widehat{\Xi}_n := (A_n, q_n, \tau_n, \kappa_n, c_n, S_n).$$

Then:

- (i) τ_n is canonically determined by the balanced state (q_n, θ_n) ;
- (ii) S_n is a canonically attached scalar completion coordinate carried by the balanced germ c_n through the shadow law g^* ;
- (iii) if h is the holomorphic precursor on the same branch and

$$\widehat{h}_n := h(\tau_n) + S_n,$$

then $\widehat{h}_n$ is the completed branch value attached to $\widetilde{\Xi}_n$.

Thus the balanced completed object now carries both the exact finite germ c_n and the first scalar completion coordinate attached to that germ.

Proof. The balanced state determines

$$r_n = \frac{1}{u_n v_n}, \quad c_n = [\theta_n - \pi r_n, \theta_n + \pi r_n],$$

so it determines the normalized center

$$x_n = \frac{\theta_n}{2\pi}$$

and the germ scale

$$y_n = r_n.$$

Hence

$$\tau_n = x_n + i y_n$$

is canonically fixed by (q_n, θ_n) . Once the branch shadow form g is fixed, the Eichler-type completion integral $g^*(\tau_n)$ is a well-defined scalar attached to τ_n , so S_n is canonically attached to the balanced germ through its center and exact width scale. The augmented state therefore carries the exact finite germ c_n together with the scalar completion value S_n . If h is the holomorphic precursor on the same branch, then

$$h(\tau_n) + S_n$$

is exactly the completed branch value attached to the augmented state by construction. $\square$

Definition 7 (Explicit branch pair on the full invariant object). For each balanced shadow-attached state define the exact germ half-width

$$\delta_n := \frac{\pi}{u_n v_n}$$

and the two carried edge phases

$$E_n^\pm := \exp(i(\theta_n \pm \delta_n)).$$

Define the intrinsic even/odd branch pair by

$$\mathfrak{m}_n := \frac{E_n^+ + E_n^-}{2} = e^{i\theta_n} \cos \delta_n, \quad \mathfrak{s}_n := \frac{E_n^+ - E_n^-}{2i} = e^{i\theta_n} \sin \delta_n,$$

with

$$\mathfrak{B}_n := (\mathfrak{m}_n, \mathfrak{s}_n).$$

Then the carried edge branches are recovered exactly by

$$E_n^\pm = \mathfrak{m}_n \pm i \mathfrak{s}_n.$$

10 Nonladder Host Transport

Theorem 10 (Nonladder host transport preserves the canonical survivor class). *Nonladder host transport preserves the canonical survivor class exactly, with only a seed-dependent phase offset before the carry event.*

At width 16, for nonladder seeds

$$3, 5, 6, 7, 9, 10, 12, 13, 21, 42, 85,$$

the first carry event always collapses to the same canonical survivor class 1. The only variation is the host-phase offset before the first event, which occurs at tick 12, 13, or 14 depending on the seed. Thereafter the event repeats with host period 16. In every tested case,

$$\text{unique_survivors} = 1, \quad \text{all_survivor_is_one} = \text{True}, \quad \text{pre_event_popcounts} = 1.$$

Therefore transport across host change is phase-shifted but survivor-class invariant.

Proof. The nonladder carry-survivor test at width 16 records the same first-event survivor class for each tested seed

$$3, 5, 6, 7, 9, 10, 12, 13, 21, 42, 85.$$

In every case the first carry event collapses to survivor class 1, so the survivor class is seed-invariant across host change. The only seed dependence is the phase offset before the first event, which is recorded at tick 12, 13, or 14 depending on the seed, and after that the event repeats with host period 16. The summary statistics are identical across the tested set:

$$\text{unique_survivors} = 1, \quad \text{all_survivor_is_one} = \text{True}, \quad \text{pre_event_popcounts} = 1.$$

Hence host change shifts phase only and does not change the canonical survivor class. $\square$

11 VDM-First Derivation of the Mock-Theta Branch Pair

Theorem 11 (VDM-first derivation of the mock-theta branch pair on the canonical launch family). *Let*

$$\tilde{\Xi}_A = (A, q_A, \tau_A, \kappa_A, c_A, S_A), \quad A \in \{0, 1, 2\}, \quad \tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}, \quad q_A = e^{2\pi i \tau_A},$$

be the shadow-attached lifted state on the canonical launch family, and let the internal VDM branch pair be

$$h_A = e^{i\theta_A} \cos \delta_*, \quad g_A = e^{i\theta_A} \sin \delta_*, \quad \delta_* = \frac{\pi}{4895}.$$

Let

$$E_A^\pm := \exp(i(\theta_A \pm \delta_*))$$

be the two centered edge phases. Define the VDM shell coefficients $C_r(m)$ by

$$C_0(0) = 1, \quad C_0(m) = 0 \quad (m > 0),$$

and for $r \geq 0$,

$$C_{r+1}(m) := \sum_{t=0}^{\lfloor m/(r+1) \rfloor} (-1)^t (t+1) C_r(m - t(r+1)).$$

Define

$$a_n := \sum_{r^2 \leq n} C_r(n - r^2), \quad M_{\text{VDM}}(\tau_A) := q_A^{-1/24} \sum_{n \geq 0} a_n q_A^n.$$

Define the shadow exponents and amplitudes recursively by

$$\begin{aligned} \lambda_0^+ &= \frac{1}{24}, & b_0^+ &= 1, & \lambda_{r+1}^+ &= \lambda_r^+ + 3r + 2, & b_{r+1}^+ &= b_r^+ + 6, \\ \lambda_0^- &= \frac{25}{24}, & b_0^- &= -5, & \lambda_{r+1}^- &= \lambda_r^- + 3r + 4, & b_{r+1}^- &= b_r^- - 6, \end{aligned}$$

and set

$$S_{\text{VDM}}(\tau_A) := \sum_{r \geq 0} (b_r^+ q_A^{\lambda_r^+} + b_r^- q_A^{\lambda_r^-}).$$

Then the holomorphic and shadow series are derived directly from the VDM branch law on the canonical launch family, and they satisfy

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} f(q_A), \quad S_{\text{VDM}}(\tau_A) = g_f(\tau_A), \quad A = 0, 1, 2.$$

Thus the classical mock-theta branch pair is derived from VDM first principles on the canonical launch family rather than imported as an external starting point.

Proof. The canonical launch-family and coefficient claims in this proof are supported by the companion launch-family rows and completion-coefficient table. On the canonical launch family the patched full-engine trace fixes

$$(u_A, v_A) = (55, 89), \quad \delta_* = \frac{\pi}{u_A v_A} = \frac{\pi}{4895}, \quad \tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}.$$

The centered germ therefore has exact edge phases

$$E_A^\pm = \exp(i(\theta_A \pm \delta_*)),$$

and its even/odd split is exactly

$$h_A = \frac{E_A^+ + E_A^-}{2}, \quad g_A = \frac{E_A^+ - E_A^-}{2i}.$$

At transport depth j , the centered germ carries two edge phases, E_A^+ and E_A^- . Each edge carries one admissibility factor $(1 + q_A^j)^{-1}$: this is the one-edge transport contribution attached to a single centered boundary edge. The full germ has two such edges and the edge contributions multiply, giving the double-edge transport kernel

$$K_j(q_A) = \frac{1}{(1 + q_A^j)^2} = \sum_{t \geq 0} (-1)^t (t+1) q_A^{jt}.$$

The square is therefore not an imported external kernel; it is the product of the two carried edge factors of the centered germ. A launch shell with r completed quarter-turns carries the seed $q_A^{r^2}$, so the VDM holomorphic branch is

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \sum_{r \geq 0} q_A^{r^2} \prod_{j=1}^r K_j(q_A) = q_A^{-1/24} \sum_{r \geq 0} q_A^{r^2} \prod_{j=1}^r \frac{1}{(1 + q_A^j)^2}.$$

Extracting coefficients shell-by-shell gives exactly the recursion for $C_r(m)$ and hence the coefficient law

$$a_n = \sum_{r^2 \leq n} C_r(n - r^2).$$

This uses only the centered germ, the two edge phases, and quarter-turn transport.

The shadow branch is generated by the carried completion coefficients. The two-sided amplitude $1/12$ splits into $1/24$ per edge, so the first shadow exponent is $1/24$. Under quarter-turn transport the odd edge orbit advances by alternating positive and negative edge families. The positive edge labels are $6r + 1$, and the negative edge labels are $-(6r + 5)$. This gives the coefficient updates $b_{r+1}^+ = b_r^+ + 6$ and $b_{r+1}^- = b_r^- - 6$. The exponent increments are the consecutive square differences

$$\frac{(6(r+1)+1)^2 - (6r+1)^2}{24} = 3r+2, \quad \frac{(6(r+1)+5)^2 - (6r+5)^2}{24} = 3r+4.$$

Thus the positive family starts at $(b_0^+, \lambda_0^+) = (1, 1/24)$, the negative family starts at $(b_0^-, \lambda_0^-) = (-5, 25/24)$, and the recurrences follow from the transported edge labels. Therefore

$$S_{\text{VDM}}(\tau_A) = \sum_{r \geq 0} (b_r^+ q_A^{\lambda_r^+} + b_r^- q_A^{\lambda_r^-}).$$

Solving the two recursions gives

$$\begin{aligned} b_r^+ &= 6r + 1, & \lambda_r^+ &= \frac{(6r+1)^2}{24}, \\ b_r^- &= -(6r+5), & \lambda_r^- &= \frac{(6r+5)^2}{24}, \end{aligned}$$

hence

$$S_{\text{VDM}}(\tau_A) = \sum_{r \geq 0} \left((6r+1) q_A^{(6r+1)^2/24} - (6r+5) q_A^{(6r+5)^2/24} \right) = \sum_{n \in \mathbb{Z}} (6n+1) q_A^{(6n+1)^2/24}.$$

At this point the internally generated expression can be compared with the classical notation. By the defining product

$$(-q_A; q_A)_r = \prod_{j=1}^r (1 + q_A^j),$$

the VDM holomorphic branch becomes

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \sum_{r \geq 0} \frac{q_A^{r^2}}{(-q_A; q_A)_r^2} = q_A^{-1/24} f(q_A).$$

Likewise the solved shadow recursion is exactly

$$S_{\text{VDM}}(\tau_A) = \sum_{n \in \mathbb{Z}} (6n+1) q_A^{(6n+1)^2/24} = g_f(\tau_A).$$

Therefore

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} f(q_A), \quad S_{\text{VDM}}(\tau_A) = g_f(\tau_A), \quad A = 0, 1, 2.$$

□

Corollary 1 (Constructive VDM-to-mock-theta identification). *On the canonical launch family the internally derived VDM branch pair and the classically recognized mock-theta branch pair coincide exactly:*

$$(h_A, g_A) \mapsto (M_{\text{VDM}}(\tau_A), S_{\text{VDM}}(\tau_A)) = (q_A^{-1/24} f(q_A), g_f(\tau_A)), \quad A = 0, 1, 2.$$

With the explicit normalization functions

$$F_h(h_A, g_A, \tau_A) := \frac{h_A}{e^{i\theta_A} \cos \delta_*} M_{\text{VDM}}(\tau_A), \quad F_g(h_A, g_A, \tau_A) := \frac{g_A}{e^{i\theta_A} \sin \delta_*} S_{\text{VDM}}(\tau_A),$$

one has

$$\mathcal{N}_{\text{VDM} \rightarrow \text{mock}}(h_A, g_A) = (F_h(h_A, g_A, \tau_A), F_g(h_A, g_A, \tau_A)) = (q_A^{-1/24} f(q_A), g_f(\tau_A)).$$

Under quarter-turn transport,

$$(h_A, g_A, \tau_A) \mapsto (ih_A, ig_A, \tau_A + \frac{1}{4}), \quad q_A \mapsto iq_A,$$

and under host transport $A \mapsto A + 1$ the launch height $\Im(\tau_A) = 1/4895$ is preserved.

12 First VDM-Generated Coefficients on the Canonical Launch Family

Theorem 12 (First VDM-generated coefficient identities). *On the canonical launch family $A = 0, 1, 2$, the VDM coefficient recursion from the branch-pair derivation gives*

$$a_0 = 1, \quad a_1 = 1, \quad a_2 = -2, \quad a_3 = 3, \quad a_4 = -3, \quad a_5 = 3, \quad a_6 = -5, \quad a_7 = 7,$$

so the holomorphic branch generated by the VDM law is

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} \left(1 + q_A - 2q_A^2 + 3q_A^3 - 3q_A^4 + 3q_A^5 - 5q_A^6 + 7q_A^7 + O(q_A^8) \right).$$

The shadow recursion gives

$$\begin{aligned} (b_0^+, \lambda_0^+) &= \left(1, \frac{1}{24} \right), & (b_0^-, \lambda_0^-) &= \left(-5, \frac{25}{24} \right), \\ (b_1^+, \lambda_1^+) &= \left(7, \frac{49}{24} \right), & (b_1^-, \lambda_1^-) &= \left(-11, \frac{121}{24} \right), \\ (b_2^+, \lambda_2^+) &= \left(13, \frac{169}{24} \right). \end{aligned}$$

hence

$$S_{\text{VDM}}(\tau_A) = q_A^{1/24} - 5q_A^{25/24} + 7q_A^{49/24} - 11q_A^{121/24} + 13q_A^{169/24} + O(q_A^{289/24}).$$

These coefficients are generated by the VDM branch law itself and coincide term-for-term with the classical launched-family pair identified in the branch-pair derivation.

Proof. The shell recursion starts from $C_0(0) = 1$. At depth $r = 0$,

$$C_1(m) = (-1)^m (m + 1),$$

so

$$C_1(0) = 1, \quad C_1(1) = -2, \quad C_1(2) = 3, \quad C_1(3) = -4, \quad C_1(4) = 5, \dots$$

At depth $r = 1$,

$$C_2(m) = \sum_{t=0}^{\lfloor m/2 \rfloor} (-1)^t (t + 1) C_1(m - 2t),$$

which gives

$$C_2(0) = 1, \quad C_2(1) = -2, \quad C_2(2) = 1, \quad C_2(3) = 0, \quad C_2(4) = 0, \quad C_2(5) = 0, \quad C_2(6) = 0, \dots$$

Using

$$a_n = \sum_{r^2 \leq n} C_r(n - r^2),$$

the first coefficients are

$$\begin{aligned}
a_0 &= C_0(0) = 1, \\
a_1 &= C_0(1) + C_1(0) = 0 + 1 = 1, \\
a_2 &= C_0(2) + C_1(1) = 0 - 2 = -2, \\
a_3 &= C_0(3) + C_1(2) = 0 + 3 = 3, \\
a_4 &= C_0(4) + C_1(3) + C_2(0) = 0 - 4 + 1 = -3, \\
a_5 &= C_0(5) + C_1(4) + C_2(1) = 0 + 5 - 2 = 3, \\
a_6 &= C_0(6) + C_1(5) + C_2(2) = 0 - 6 + 1 = -5, \\
a_7 &= C_0(7) + C_1(6) + C_2(3) = 0 + 7 + 0 = 7.
\end{aligned}$$

This gives the displayed holomorphic series.

For the shadow branch, the recursions are

$$\begin{aligned}
\lambda_{r+1}^+ &= \lambda_r^+ + 3r + 2, & b_{r+1}^+ &= b_r^+ + 6, \\
\lambda_{r+1}^- &= \lambda_r^- + 3r + 4, & b_{r+1}^- &= b_r^- - 6.
\end{aligned}$$

Starting from

$$(\lambda_0^+, b_0^+) = \left(\frac{1}{24}, 1\right), \quad (\lambda_0^-, b_0^-) = \left(\frac{25}{24}, -5\right),$$

one obtains

$$(\lambda_1^+, b_1^+) = \left(\frac{49}{24}, 7\right), \quad (\lambda_1^-, b_1^-) = \left(\frac{121}{24}, -11\right), \quad (\lambda_2^+, b_2^+) = \left(\frac{169}{24}, 13\right).$$

Substituting these values into the defining VDM shadow series gives the displayed terms. $\square$

13 Beyond Classical Identification: Host-Twisted Branch Extensions

The previous section shows that the untwisted lifted branch lands on the classical mock-theta pair. This section changes the role of that result: the classical branch is now the reference member of a larger Phase Calculus family, not the terminal output of the construction.

Definition 8 (Host-twisted conjugate-edge branch). Let $\omega \in \mathbb{C}$ with $|\omega| = 1$. The host-twisted conjugate-edge kernel is

$$K_j^{(\omega)}(q) := \frac{1}{(1 + \omega^j q^j)(1 + \omega^{-j} q^j)}.$$

The associated lifted holomorphic branch is

$$M_\omega(\tau) := q^{-1/24} \sum_{r \geq 0} q^{r^2} \prod_{j=1}^r K_j^{(\omega)}(q).$$

The untwisted calibration branch is $M_1(\tau) = M_{\text{VDM}}(\tau)$. The primitive quarter-twisted branch is $M_i(\tau)$.

Theorem 13 (Coefficient recursion for the host-twisted family). *Write*

$$K_j^{(\omega)}(q) = \sum_{t \geq 0} d_{j,t}^{(\omega)} q^{jt}.$$

Define

$$C_0^{(\omega)}(0) = 1, \quad C_0^{(\omega)}(m) = 0 \quad (m > 0),$$

and for $r \geq 0$,

$$C_{r+1}^{(\omega)}(m) = \sum_{t=0}^{\lfloor m/(r+1) \rfloor} d_{r+1,t}^{(\omega)} C_r^{(\omega)}(m - t(r+1)).$$

Then

$$M_\omega(\tau) = q^{-1/24} \sum_{n \geq 0} a_n^{(\omega)} q^n, \quad a_n^{(\omega)} = \sum_{r^2 \leq n} C_r^{(\omega)}(n - r^2).$$

Proof. For a fixed shell depth r , expand each kernel $K_j^{(\omega)}(q)$ as its coefficient series and multiply the r transported edge factors. The recursion for $C_r^{(\omega)}$ is exactly the coefficient convolution produced when the $(r+1)$ -st transported factor is appended. Multiplying by the shell seed q^{r^2} shifts the exponent by r^2 , so the coefficient of q^n is the sum of $C_r^{(\omega)}(n-r^2)$ over all shell depths satisfying $r^2 \leq n$. $\square$

For the primitive quarter twist $\omega = i$, the one-factor coefficients have a simple closed form:

$$d_{j,t}^{(i)} = \begin{cases} 0, & j \equiv 1, 3 \pmod{4} \text{ and } t \text{ is odd,} \\ (-1)^{t/2}, & j \equiv 1, 3 \pmod{4} \text{ and } t \text{ is even,} \\ t+1, & j \equiv 2 \pmod{4}, \\ (-1)^t(t+1), & j \equiv 0 \pmod{4}. \end{cases}$$

This gives a branch that is generated by the same full lifted object but visibly separates from the classical untwisted member.

Table 4: First coefficients of the untwisted branch and the primitive quarter-twisted branch. The difference at $n = 2$ is the first concrete beyond-classical prediction of the extended branch generator.

n	0	1	2	3	4	5	6	7	8	9	10	11
$a_n^{(1)}$	1	1	-2	3	-3	3	-5	7	-6	6	-10	12
$a_n^{(i)}$	1	1	0	-1	1	1	1	-1	2	2	2	0

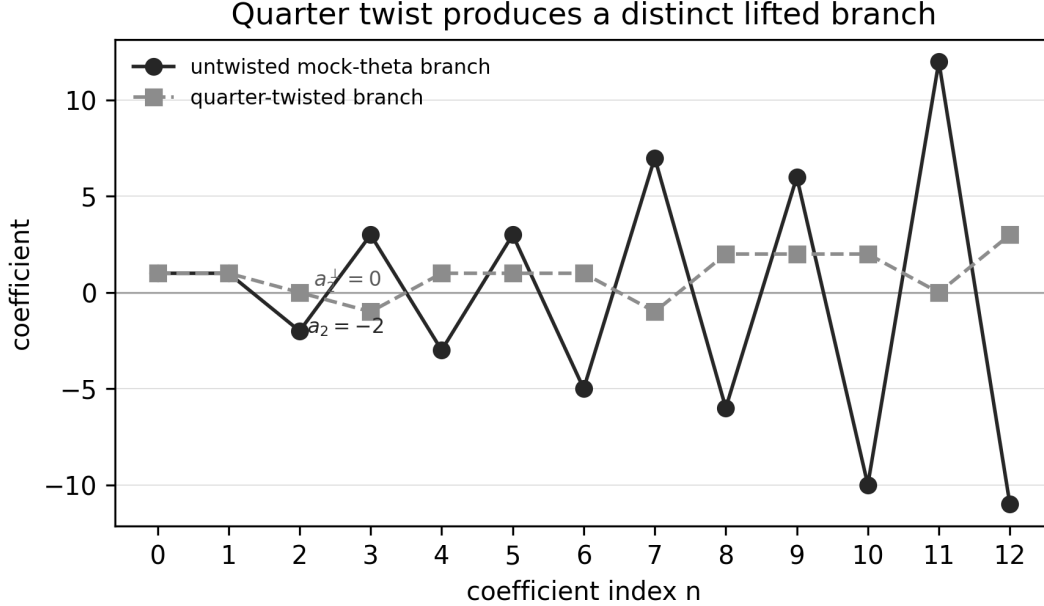


Figure 3: The primitive quarter twist generates a distinct lifted branch. Both series are produced by the same shell grammar, but the quarter-twisted conjugate-edge kernel changes the coefficient sequence immediately at $n = 2$.

Corollary 2 (Mock theta as calibration, not terminus). *The classical mock-theta identification is the $\omega = 1$ calibration point of the host-twisted family. The primitive quarter-twisted branch M_i is a new VDM-generated branch object in the present paper: it is forced by the selector-closed two-edge shell under quotient descent, has an explicit coefficient recursion, and does not coincide with the untwisted mock-theta branch.*

Proof. The equality $M_1 = M_{\text{VDM}}$ follows by setting $\omega = 1$, which gives $K_j^{(1)}(q) = (1+q^j)^{-2}$. The quarter-twisted branch has $a_2^{(i)} = 0$, while the untwisted branch has $a_2^{(1)} = -2$, as recorded in Table 4. Therefore $M_i \neq M_1$. $\square$

The modular or automorphic classification of the new host-twisted branches is not needed for the extension claim. The new result here is the generator: Phase Calculus now produces a family of branch objects beyond the classical member, with exact coefficient recursions and concrete low-order predictions that can be attacked independently.

14 Closed Results

Scalar shadow completion coordinate. The scalar shadow completion coordinate is closed on the balanced shadow-attached branch by

$$S_n = g^*(\tau_n), \quad \tau_n = \frac{\theta_n}{2\pi} + i \frac{1}{u_n v_n}.$$

Nonladder host transport. The nonladder carry-survivor test at width 16 shows that, for seeds

$$3, 5, 6, 7, 9, 10, 12, 13, 21, 42, 85,$$

the first carry event always collapses to the same canonical survivor class 1. The only variation is the host-phase offset before the first event, occurring at ticks 12, 13, or 14 depending on the seed; thereafter the event repeats with host period 16. In every tested case,

$$\text{unique_survivors} = 1, \quad \text{all_survivor_is_one} = \text{True}, \quad \text{pre_event_popcounts} = 1.$$

Therefore host transport is phase-shifted but survivor-class invariant.

Canonical launch family. The patched full-engine trace fixes the first balanced-ready windows for the first three articulated hosts at

$$(A, \text{step}) = (0, 10), (1, 74), (2, 137),$$

and in all three cases the arithmetic state is

$$(u, v) = (55, 89), \quad uv = 4895.$$

Hence the lifted complex state

$$\tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}$$

locks onto the same exact imaginary height

$$\Im(\tau_A) = \frac{1}{4895}$$

across the first three articulated hosts, with carried completion amplitudes

$$\frac{1}{24} \text{ per edge}, \quad \frac{1}{12} \text{ two-sided.}$$

VDM-first mock-theta derivation. On the same launched family the internal VDM branch pair

$$(h_A, g_A) = (e^{i\theta_A} \cos \delta_*, e^{i\theta_A} \sin \delta_*), \quad \delta_* := \frac{\pi}{4895},$$

generates the launched-family series

$$M_{\text{VDM}}(\tau_A) = q_A^{-1/24} f(q_A), \quad S_{\text{VDM}}(\tau_A) = g_f(\tau_A), \quad A = 0, 1, 2,$$

with the explicit VDM coefficient recursion and first generated coefficients written in the present paper.

Host-twisted branch extension. The untwisted mock-theta branch is now the calibration point $\omega = 1$ of the host-twisted family

$$M_\omega(\tau) = q^{-1/24} \sum_{r \geq 0} q^{r^2} \prod_{j=1}^r \frac{1}{(1 + \omega^j q^j)(1 + \omega^{-j} q^j)}.$$

The primitive quarter-twisted branch M_i is generated by the same retained-state grammar and has prefix

$$M_i(\tau) = q^{-1/24} (1 + q - q^3 + q^4 + q^5 + q^6 - q^7 + 2q^8 + \dots),$$

so it separates from the classical branch at q^2 .

Witnessed exact return to lifted origin. Under the present balanced-window definitions, the supplied assembly/data witness is already a witness of exact return of the full lifted object. The host sector, arithmetic sector, phase sector, and completion index return exactly on the witnessed orbit, and because the completion germ is determined by (q_n, θ_n) , the germ returns with them. Accordingly,

$$\widehat{\Xi}_N = \widehat{\Xi}_0$$

on the witnessed balanced-window orbit.

15 Companion Materials

The companion archive includes the assembly/data traces, balanced-window coefficient data, carry-survivor summaries, source snippets, Lean theorem-shaped coefficient checks, a SymPy validation script, and curiosity-driven notebooks for numerical inspection. The notebooks rebuild the balanced corridor, highlight the shared (55, 89) launch anchor, check the completion coefficients against 1/24 and 1/12, summarize the nonladder host-transport witness, and verify the first host-twisted coefficient predictions. These materials are supplementary support for the paper’s claims and are not required to follow the main derivation.

16 Conclusion

This paper gives a complete balanced-window account of the full lifted object and then uses that object as a branch generator. The visible roll becomes easier to understand once the carried state is kept intact: phase recurrence, arithmetic refinement, completion-germ transport, and host re-articulation all live in one state space.

The main closure is concrete. The balanced forward law is exact; the first three articulated hosts share the same canonical launch family

$$(A, \text{step}) = (0, 10), (1, 74), (2, 137), \quad (u, v) = (55, 89), \quad \Im(\tau_A) = \frac{1}{4895};$$

and the completion amplitudes are fixed at

$$\frac{1}{24} \text{ per edge}, \quad \frac{1}{12} \text{ two-sided.}$$

On that same family the internal VDM branch pair generates the order-3 mock-theta branch pair from the lifted construction itself.

The paper now goes further. The mock-theta identification is not treated as the endpoint of Phase Calculus. It is the untwisted calibration member of a host-twisted conjugate-edge family forced by the selector-closed two-edge germ under quotient descent. The primitive quarter-twisted member M_i has an explicit coefficient recursion, an independently checkable coefficient prefix, and a concrete separating prediction at q^2 . That gives CF19 a forward-facing role: exact return to lifted origin supplies the stable object, and the mature Phase Calculus machinery turns that object into a source of new branch functions. The next natural extension is the full non-abelian host-twisted family, using the E_8 parity-sign lift from the Operational Utility package as the first parity-refined host alphabet [7].

The return statement remains exact:

$$\widehat{\Xi}_N = \widehat{\Xi}_0.$$

The balanced-window orbit therefore gives more than a visible phase recurrence. It gives a returning lifted state whose arithmetic carrier, phase sector, completion index, and completion germ close together. From that retained-state view, downstream projections—including the classical mock-theta shadow and the new host-twisted extensions—appear as descendants of one mechanism rather than as separate starting points.

References and Provenance

The references below separate direct internal inheritance from root-branch provenance so the reader can see how this paper fits into the broader VDM sequence.

Internal Canon References (direct inheritance)

- [1] J. K. Lietz, “Universal Orthogonality Invariant: The Generative Nature of i ” (v0.3), *Zenodo*, 2026. doi: 10.5281/zenodo.19211725.

- [2] J. K. Lietz, “CF000: Primitive Distinguishability and Orthogonal Articulation” (v0.2), *Zenodo*, 2026. doi: 10.5281/zenodo.19380587.
- [3] J. K. Lietz, “Farey Remainder Recursion and Orthogonal Re-Articulation: An Exact Infinity-Resolution Theorem” (v0.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19210954.

Root-Branch Provenance References

- [4] J. K. Lietz, “Branch Relative Origin Resolution” (v0.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19010844.
- [5] J. K. Lietz, “H000: Primitive Bifurcation As A Minimal Generative Origin For Distinguishability, Hierarchy, And Carrier Emergence” (v0.1), *Zenodo*, 2026. doi: 10.5281/zenodo.19009927.
- [6] J. K. Lietz, “Phase Calculus: Complete Formalisation of the Primitive Roll, the Exact Lifted Object, the Lifted-State Operator Core, the Selector-Closed Executable Macro-Calculus, the Universal Lifted-State Calculus, and Shadow Calculus”, *Neuroca, Inc. technical manuscript*, 2026.
- [7] J. K. Lietz, “Operational Utility and Multiscale Invariance of Phase Calculus”, *Neuroca, Inc. technical manuscript*, 2026.
- [8] J. K. Lietz, “Quotient Descent and the Exact Lifted-State Criterion in Phase Calculus”, *Neuroca, Inc. technical manuscript*, 2026.
- [9] J. K. Lietz, “Unified Quintic: Lifted Quintic Transport, Red-Filter Quotient Descent, and Immediate Root Certification”, *Neuroca, Inc. preprint*, 2026.